



DRESDEN LEIPZIG

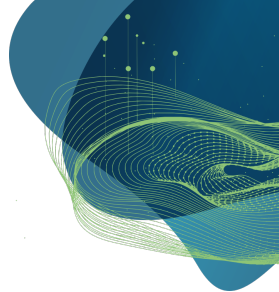
Center for Scalable Data Analytics and Artificial Intelligence

Digitization and Data Analytics

Consequences

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GEFÖRDERT VOM



Bundesministerium
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und Forschung

SACHSEN



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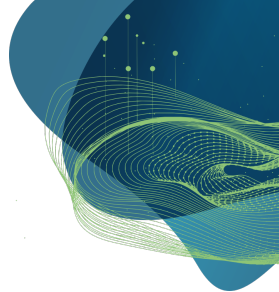


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Agenda

- 1 Recap and Introduction
- 2 Data, Methods, Infrastructure
- 3 Non-Technical Issues
- 4 Conclusions



1 Recap and Introduction – Agenda

- 1 Recap and Introduction
 - Hopes, Promises and Visions on AI
 - Nightmares, Warnings and Dystopias of AI
- 2 Data, Methods, Infrastructure
- 3 Non-Technical Issues
- 4 Conclusions

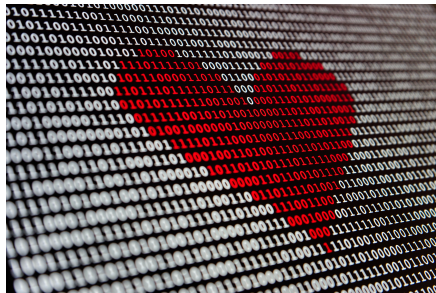
Recap and Introduction

- Short recap of the topics:
 - ▶ Data: Data Management, Data Preprocessing
 - ▶ Methods: Statistics, Unsupervised Learning, Supervised Learning, Deep Learning
 - ▶ Hardware/Tools: Distributed Computing, Distributed Machine Learning
- This lecture: provide a broader perspective on AI and its implications
- Many more aspects than covered here $\leftrightarrow$ depends on individual perspective
- Much of this lecture: highly subjective

For the sake of simplicity, the terms data analytics, machine learning, and artificial intelligence will be collectively referred to as AI in the following.

Hopes, Promises and Visions on AI

- Improving efficiency and productivity
- Automation of mundane tasks
- Enhanced decision-making through data-driven insights
- Mitigation of societal challenges
- Breakthroughs in scientific research and discovery
- Revolution of transportation systems through autonomous vehicles
- ...



What are your visions and hopes on AI?

Nightmares, Warnings and Dystopias of AI

- Job displacement and automation
- Privacy and security
- Lack of transparency and explainability
- Ethical dilemmas
- Dependency and reliability
- ...



https://www.freepik.com/free-ai-image/drawing-skull-with-bunch-plants-it_40570441.htm

What are your fears and nightmares on AI?

2 Data, Methods, Infrastructure – Agenda

- 1 Recap and Introduction
- 2 Data, Methods, Infrastructure
 - Data
 - Methods
 - Target Metrics
 - Processing, Computing, and Infrastructure
- 3 Non-Technical Issues
- 4 Conclusions

Data

- Successful application of whatever method is mainly about data quality
- Data cleaning and preprocessing is essential here and takes most of the time, but is quite boring and not very appealing
- Documentation: what is the data about?, check some examples from **kaggle.com**:
 - ▶ Popular dataset I: <https://www.kaggle.com/datasets/shivamb/netflix-shows>:
Hint: Provided suggestions for interesting analysis tasks. Who defines your analysis goal?
 - ▶ Popular dataset II:
<https://www.kaggle.com/datasets/antgoldbloom/covid19-data-from-john-hopkins-university>
How to define 'corona death'? What other data are needed to get valuable insights? (cf. off-statistical issues from lecture Statistics)
- Learning from Kaggle datasets beneficial wrt. applying methods etc. – but only about secondary data
- Primary data from planned experiments provides far more insights
- Simply taking data from somewhere (e.g. Kaggle) strongly misleading wrt. effort for data generation and application of data analysis

Methods

Method selection should be based on

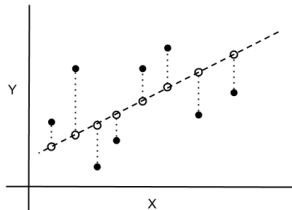
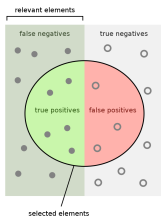
1. Problem understanding, **target**:
 - ▶ **Understanding** context, scope, desired outcomes
 - ▶ **Explore** patterns, make predictions, uncover associations
2. Scalability and efficiency:
 - ▶ $\leftrightarrow$ dealing with large datasets
 - ▶ Some **methods** may be computationally intensive or may not **scale** well with increasing data size
3. Assumptions and limitations:
 - ▶ Determines interpreting results accurately and avoiding misinterpretations or biased conclusions
4. Interpretability and explainability:
 - ▶ Depending on application these aspects may be crucial
 - ▶ Decision trees or linear regression offer **interpretable** models, while others, e.g. deep learning algorithms, provide less explainability
 - ▶ Important, esp. in regulated domains or when human decision-making is involved

Consider Wolpert's No-Free-Lunch Theorem, cf. <https://doi.org/10.1109/4235.585893>:

There is no single method that performs universally well on all problems.

Target Metrics

- Different target metrics:
 - ▶ Classification: accuracy, precision, recall, f1 score, ...
 - ▶ Mean absolute error, mean squared error, R^2 , ...
- Every metric has its own characteristics



- Not all metrics can be optimized to max or min simultaneously
- Depending on application: decide for most important metric(s)



Target Metrics

Medical Diagnosis

- Diagnosing diseases or medical conditions based on patient symptoms and test results
- Target metric(s): sensitivity and specificity
Prevent system from not identifying relevant cases (sensitivity, true positive rate) and prevent healthy patients from being treated accidentally (specificity, true negative rate)

Autonomous driving

- Critical decisions as object detection and collision avoidance
- Metric: False Negative Rate (FNR)
Minimize rate at which system fails to identify a potential danger or obstacle

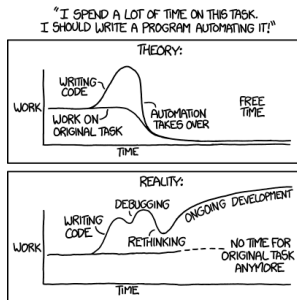
Financial Risk Assessment

- Quantifying risk associated with investments (e.g. credit default risk)
- Metric: Value at Risk
Quantifies potential loss that is not exceeded given a predefined probability, e.g. 99.9% (over a specified time horizon)

Processing, Computing, and Infrastructure

- Establishing appropriate infrastructure is crucial for effective data processing and analysis
- Hardware **costs** are decreasing (CPUs, GPUs, memory, storage)
- Virtualization and cloud computing make strong infrastructure more **easily accessible**
- Broad and well developed ecosystem(s) available esp. for software stack (tools, frameworks)
- Recent challenges esp. for distributed systems:
 - ▶ I/O: bring **data close to computation**
 - ▶ **Distribution** and **parallelization** strategies (e.g. large language models)

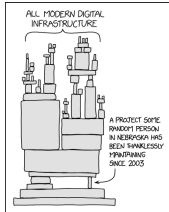
- Theory and reality of automation:



<https://xkcd.com/1319>

Processing, Computing, and Infrastructure

- Advanced technology alone is not sufficient: need for high-quality data and domain knowledge
 - Complex systems require specialized knowledge and expertise
 - Main challenge for complex problems: integrating data, hardware and tools
 - Reliability, stability, maintainability, ...
- Human factor: adoption of new technologies (hardware, software, processes, ...) heavily depends on human factor, e.g.
 - ▶ Introducing new standard file format for production data across long product chain
 - ▶ Convince group of people to use some special tool/software



<https://xkcd.com/2347>

3 Non-Technical Issues – Agenda

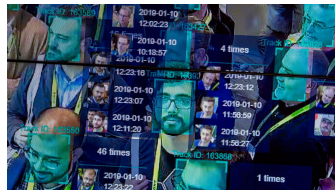
- 1 Recap and Introduction
- 2 Data, Methods, Infrastructure
- 3 Non-Technical Issues
 - AI Applications
 - Reproducibility and Data Access
 - Deepfakes
 - Astroturfing
 - Cybersecurity
 - Ethical AI
 - Legal Aspects
 - Resource-Related Issues
 - Geopolitical Dimension
 - AI in Practice: Roles and Responsibilities
- 4 Conclusions

Non-Technical Issues

- What if AI learns from AI?
- Training data esp. for LLMs: internet content AI generated → training data for LLMs taken from the internet → AI learns from itself
- Directly create e.g. text data with LLMs
- AI annotated training data (research: annotation of risk for traffic situation by humans vs. by LLM)
- AI driven data-generation might reproduce intentionally implemented aspects

AI Applications

- Facial recognition example: <https://news.sky.com/story/met-polices-facial-recognition-tech-has-81-error-rate-independent-report-says-11755941>
- Metropolitan's police (responsible for area of Greater London, UK) has been monitoring crowds with live facial recognition since August 2016 at Notting Hill Carnival
- Conducted 10 further trials at locations including Leicester Square, Westfield Stratford, and Whitehall during the 2017 Remembrance Sunday commemorations
- Researchers found that system is 81% inaccurate, majority of cases, it flagged up faces to police when they were not on a wanted list
- Claim: technology only makes mistake in one in 1,000 cases
- Cp. example from lecture Statistics (chance findings, base rate fallacy, face recognition): quite similar number of 80% fails



Reproducibility and Data Access

- “Data hugging”: proprietary datasets prevent independent verification of AI claims <https://doi.org/10.1038/s44387-026-00094-2>
- Systematic review of approx. 1300 critical-care AI studies:
 - ▶ 85% did not share datasets, 87% did not share code
- Case study: Apple claims biological age prediction from wrist PPG with MAE $\approx$ 2.89 years
- Independent replication on UK Biobank (Karpurapu et al. 2026): MAE $\approx$ 5.24–5.97 years – only marginally better than predicting the mean (6.98 years)
- Real consequences of unverified deployment:
 - ▶ Epic sepsis algorithm: poor external validation *after* nationwide deployment
 - ▶ Philips ECG telemetry: 2 deaths, 109 injuries from missed alerts (FDA Class I correction)



If you cannot reproduce it, you cannot trust it – regardless of how good the ML is.

Deepfakes

- Deepfakes might influence elections (Nov. 2023):

<https://business.outlookindia.com/technology/>

deepfake-elections-how-indian-politicians-are-using-ai-manipulated-media-to-malign-opponents

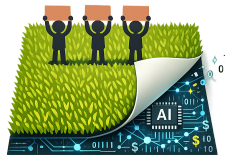
- Deepfake videos being used to manipulate voters' perceptions in India (minister appealing to vote against the present state government, quiz show asking questions around Madhya Pradesh politics to whip up anti-incumbency sentiments among viewers)
- Political parties employing private-sector advocacy and campaign agencies to create and distribute deepfake videos
- Used to spread misinformation about candidates
- Even established political parties not fighting against deepfakes because using deepfakes themselves to gain advantage



What are the implications for democracy?

Astroturfing

- Corporate astroturfing is not new — but AI has made it radically cheap
- **Alliance of Australian Retailers (2010):** 19,000 “small business” members mobilized to fight Australia’s world-first plain cigarette packaging law <https://www.tobaccotactics.org/article/alliance-of-australian-retailers/>
- Presented as an authentic grassroots coalition of corner stores, petrol stations and newsagents, millions of dollars in campaign funding
- But: Philip Morris International, British American Tobacco and Imperial Tobacco secretly financed the AAR
- **The shift with AI:** what cost millions per year now costs €10/month + one afternoon using an LLM and an automation tool



Cybersecurity

- Same technology, adversarial direction: AI agents change the security landscape
- Fully autonomous attacks without human intervention
 - ▶ Adaptive attacks at machine speed – no human latency
- Attacks on agents themselves (e.g. prompt injection)
 - ▶ AI agents are “digital insiders”: privileged entities with access, autonomy, and inherited trust



Attacker–defender asymmetry has shifted sharply: attacks scale with compute, defenses still scale with headcount!

Ethical AI

- What is ethics?
 - ▶ Ethics is not about assessing **moral** status of actions
 - ▶ Very roughly: ethics tries to identify relevant societal **implications** while using an AI system
 - ▶ Some main perspectives in ethics: **deontological** (assumes an absolute moral status of some action), **consequentialism** (moral status of an action depends on consequences), **descriptive/empirical** (moral status is derived empirical by observing/asking people)
- How to implement ethical AI?
- *question to the audience*: Which ethical perspective to implement?
- Last years: many suggestions of '**ethical**' **frameworks** came up to consider ethical aspects while development, deployment, usage of AI systems
 - ▶ Approaches of 'certifying' AI systems
 - ▶ E.g. Value-based Engineering with ISO/IEC/IEEE 24748-7000 (see <https://www.wu.ac.at/value-based-engineering/principles/>)
 - ▶ Ecosystem **Responsibility**, Stakeholder Inclusiveness, Context-Sensitivity, Value Identification with Moral Philosophy and/or Spiritual Tradition, Understanding values at depth, Leadership Engagement, Respect for Regional Laws and International Agreements, Willingness to renounce Investment, Transparency of the Value Mission, Risk-based System Design

Application I of Ethics in AI

- US justice applies COMPAS¹—a proprietary ML model—for recidivism prediction:
<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
 - ▶ COMPAS as proprietary model with unknown decision process behind in the background (strong relation to explainability of AI, cf. <https://doi.org/10.1038/2Fs42256-019-0048-x>)
 - ▶ Criminologists accused COMPAS being racially biased
 - ▶ Typos in input data for COMPAS (130+ factors) can lead to unfairness
- Some ethical questions:
 - ▶ Should we rely on machine-driven decisions for high-stake decisions at all?
 - ▶ Is it really intended to have some kind of 'objectivity'? Esp. subjective decisions that consider the individual can be more appropriate.
 - ▶ How can we measure (un)fairness or bias?

¹Correctional Offender Management Profiling for Alternative Sanctions

Application II of Ethics in AI

- Evaluating toxicity² in texts by extracting 'moral scores' in natural language processing from BERT model:
<https://doi.org/10.48550/arXiv.2103.11790>
 - ▶ Moral score is based on many assumptions and abstractions, e.g. using 1st PC, normalization based on maximum raw score, survey data for calibration (whom to ask, what to ask, ...)
 - ▶ Moral score in interval [-1,1], defining $ms < -0.5$ as 'toxic'
- Some follow-up questions:
 - ▶ Why is the threshold at -0.5? What are interpretable drivers?
 - ▶ How can the difference between, e.g. -0.4 and -0.5 be interpreted?
 - ▶ The approach relies on deontological ethics (as it is rule-based easy to implement from algorithmic perspective). Is that somehow societal consensus? Who defines/decides about good and bad?

²Note that 'toxicity' wrt. to texts is a quite new term/phenomenon coming up during the last years.

Application III of Ethics in AI

Examples **toxicity** of texts, screenshots taken from <https://www.perspectiveapi.com/case-studies/> (own highlighting):

Southeast Missourian

Improving conversations and building community via online comments

Local newspapers help people learn about and act on local issues. The forums of local news outlets have taken the place of town squares, and offer communities the opportunity for genuine connection and local participation.

The Southeast Missourian was ready to turn off the comment section because it was too expensive and time consuming to moderate. Instead, they integrated with Perspective API to help manage the conversations happening on their site. Perspective API gives a toxicity score to each piece of content. Using this score as a guide, Southeast Missourian moderators can make the final decision to remove or not allow a comment to be posted. Due to this experiment, comments that were highly likely to be toxic were reduced by 96%. Now, this digital town hall can foster a sense of community without fear of harassment.

Open Web

Moderating comments in real-time to improve conversations online.

OpenWeb is a leading audience engagement platform that hosts 100 million active users per month. Their clients include publishers like AOL, HuffPost, TechCrunch, Fox News, Hearst, Refinery29, Sky Sports, MarketWatch, Salon, Newsweek and many others. Their mission is to elevate conversations online: they want to inspire an open exchange of ideas across the web — with less hostility and toxicity. Their goal is to encourage healthy debate and make room for diverse opinions, without suppressing free speech.

OpenWeb's moderation platform has been using Jigsaw's Perspective API for a little over three years but they recently launched a "Real-Time Feedback" feature with the API at its core. The feature moderates comments in real-time, using Perspective API to determine whether comments violate community guidelines. The user can either choose to edit their comment and repost, or they can post it anyway, accepting the outcome.

This new feature was launched after an OpenWeb study, conducted as part of a broader Jigsaw research study, found that out of all commenters who edited their language, 44.7% of commenters replaced or removed offensive words, and 7.6% elected to rewrite their comment entirely.

- AI used as a decision supporting system for moderators
- humans are typically **lazy**: using simply the machine's decision
- **responsibility**: maybe easier to justify a decision based on some machine score but not on some individual reasoning

- Assistant system for users, provide direct reflection wrt. toxicity
- Does such reflection influences long term behavior and thinking?
- Is it indeed a good idea to rewrite/reformulate some comment based on this? (**censorship** works mainly the same way)
- At the very end: making only some algorithm happy?

Application III of Ethics in AI

Issues and consequences of computer-aided or automated decisions:

- How to justify decisions that are in contradiction to the algorithm?
- Does human laziness influences the decision?
- What about the impact on trust in human decision-making ability and innovation capacity?
- What about responsibility and liability for consequences of decisions?
- What happens to the human ability to make decisions?

Legal Aspects

- Who is legally responsible for wrong or misleading AI output?
- Air Canada chatbot invented a refund policy on the airline's website
- Tribunal ruling (Feb 2024): Air Canada must honor the invented policy

<https://arstechnica.com/tech-policy/2024/02/>

[air-canada-must-honor-refund-policy-invented-by-airlines-chatbot/](#)



Legal Aspects

- Microsoft Services Agreement, section 3 (see <https://www.microsoft.com/en/servicesagreement>): Code of Conduct enforced by AI
 - 4.b.iii. Review. Where applicable, *we may use automated systems and humans* to review content to identify [...] illegal or harmful content or conduct.
- What is *harmful* content? → no legal definition
- Common practice at Google, Apple, Meta, ...
- Suspicious scan results forwarded to authorities, police investigations followed³
- Accounts suspended even after cases were closed – no reactivation, complete loss of data
- Bottom line: the risk sits on the customer's side, not the platform's
- Under the EU AI Act (Aug 2026), providers must at least *disclose* that automated systems are involved (disclosure is not remedy!)

³ see <https://www.heise.de/hintergrund/Automatisierte-Scans-Microsoft-sperrt-Kunden-unangekündigt-für-immer-aus-7324608.html>,
<https://www.nytimes.com/2022/08/21/technology/google-surveillance-toddler-photo.html>

Legal Aspects

- EU AI Act: in force since 1 August 2024 – first comprehensive AI regulation worldwide
- Risk-based approach: unacceptable / high / limited / minimal risk
- Phased applicability (as of July 2026):
 - ▶ Since Feb 2025: prohibitions on unacceptable-risk systems (social scoring, cognitive manipulation, real-time remote biometric ID)
 - ▶ Since Aug 2025: obligations for General-Purpose AI (GPAI) models
 - ▶ From 2 August 2026, transparency: chatbots must disclose AI, deepfakes must be labeled as artificially generated
 - ▶ High-risk system obligations phased in through 2027
- General exception: military, defence, national security
- Details: <https://artificialintelligenceact.eu/>

Resource-Related Issues

Development and application of AI is a lot about resources.

- Resources:
 - ▶ Financial
 - ▶ People/knowledge
 - ▶ Time
 - ▶ Technical/hardware
- Some numbers⁴
 - ▶ OpenAI started with a budget of 1 billion USD in 2015, further capital by Microsoft⁵ in 2019 (1 billion USD) and 2023 (10 billion USD)
 - ▶ Nvidia invested 100 million USD in 2024,
<https://www.investors.com/news/technology/nvidia-stock-microsoft-stock-openai-funding-valuation>
 - ▶ but: OpenAI not profitable (loss 5 billion USD in 2024,
<https://www.cnn.com/2024/09/27/openai-sees-5-billion-loss-this-year-on-3point7-billion-in-revenue.html>)
 - ▶ Mid 2018: GPT-1 (approx. 2-3 years development time)
 - ▶ May 2020: GPT-3 (32 authors on preprint article, <https://arxiv.org/abs/2005.14165>)
 - ▶ March 2023: GPT-4 (284 authors on preprint article, <https://arxiv.org/abs/2303.08774>)
 - ▶ OpenAI number of employees (<https://seo.ai/blog/how-many-people-work-at-openai>): Nov. 2023: 770, Sept. 2024: $\approx$ 3500, by end of 2026: up to 8000⁶

⁴<https://blog.gitnux.com/chat-gpt-statistics/>, <https://klu.ai/blog/gpt-4-llm>

⁵MSFT (sales, net profit) 2019: (125 billion USD, 39 billion USD), 2023: (212 billion USD, 72 billion USD).
<https://www.google.com/finance/quote/MSFT:NASDAQ?sa=X&ved=2ahUKEwisgdaEzIiAAxU7TaQEHTeEAVAQ3ecFegQIRBAZ>

⁶<https://www.reuters.com/business/openai-nearly-double-workforce-8000-by-end-2026-ft-reports-2026-03-21/>

Resource-Related Issues

Model (Year)	Parameters	Training Tokens	Accelerators	Training Compute
GPT-3 ⁷ (OpenAI, 2020)	175 B	300 B	1k Nvidia V100	800k GPU-hours (34 days)
GPT-4 ⁸ (OpenAI, 2023)	1.8 T	13 T	25k Nvidia A100	54 M GPU-hours (90–100 days)
Llama 3.1 ⁹ (Meta, 2024)	405 B	15 T	16k Nvidia H100	30 M GPU-hours

- 750 words $\approx$ 1,000 tokens
- Cloud pricing (on-demand): Nvidia H100 $\sim$ USD 2–7 / GPU-hour
- ZIH GPU cluster *Capella* at TU Dresden: 624 Nvidia H100 GPUs

⁷ <https://www.gpunex.com/blog/ai-training-costs-2026/>

⁸ see <https://klu.ai/blog/gpt-4-llm>

⁹ <https://huggingface.co/meta-llama/Llama-3.1-405B>

Geopolitical Dimension

- Once resources of this **scale** are in play, frontier AI stops being a product decision
- Case: Anthropic's Claude "Mythos" release, June 2026
- White House restrictions on foreign access based on **export-control regulations**

https:

[//www.businessinsider.com/anthropic-white-house-fable-mythos-5-drama-explained-2026-6](https://www.businessinsider.com/anthropic-white-house-fable-mythos-5-drama-explained-2026-6)

- First time a **commercial LLM release triggered an export-control**-style response within a news cycle
- Frontier models are now dual-use technology
 - ▶ Capability thresholds trigger government intervention
 - ▶ AI power is no longer just economic, it is **geopolitical**



AI is no longer only a product decision.

AI in Practice: Roles and Responsibilities

Topic	Job Label(s)	Scope of Work	Typical Task / Question
Introduction	Data Strategy Consultant	 Research	<i>"Where in our organization can data create value, and what should we tackle first?"</i>
Data Preprocessing	Data Engineer	 Production	<i>"How do I clean, transform, and pipeline this raw data so it's reliably usable downstream?"</i>
Statistics	Data Analyst	 Interpretation	<i>"What does the distribution of customer churn look like last quarter, and what trends stand out?"</i>
Data Management	Data Steward	 Research	<i>"How do we document, version, and store this dataset so it stays reproducible and reusable in 10 years?"</i>
Machine Learning & Deep Learning	• Data Scientist • Machine Learning Engineer • AI Research Engineer	 Interpretation +  Production +  Research	• "Are there hidden customer segments or anomalies in our data?" (Data Scientist) • "Can I train and deploy a model – classical or neural – that meets our accuracy and latency targets?" (ML Engineer) • "Can a novel architecture or training method outperform the current state of the art?" (AI Research Engineer)
Distributed Computing	• Big Data Engineer • MLOps Engineer	 Production	• "How do I move and process 50 TB of streaming data per day with Spark/Kafka?" (Big Data Engineer) • "Is my deployed model still accurate – and how do I retrain, monitor, and roll it back safely?" (MLOps Engineer)
Consequences	AI Ethics Officer	 Interpretation (societal & ethical reflection)	<i>"Does this model treat all user groups fairly, and what are the legal and social implications of deploying it?"</i>

4 Conclusions – Agenda

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Conclusions

- AI is not only about methods and hardware
- Using AI systems might have large implications on human behavior and society
- Distinction between high-stake and low-stake decisions
- Many aspects are hard to cover/measure, as e.g. ethics or legal aspects
- Many further aspects that are touched as well, were not covered here: esp. economic aspects
- The bottom line: The most relevant issues and questions while using AI broadly are related to non-technical questions.

